International Journal of Child-Computer Interaction 28 (2021) 100234

Contents lists available at ScienceDirect

International Journal of Child-Computer Interaction

journal homepage: www.elsevier.com/locate/ijcci

A social robot's influence on children's figural creativity during gameplay

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ARTICLE INFO

Article history:
Received 1 September 2020
Received in revised form 19 November 2020
Accepted 20 November 2020
Available online 2 December 2020

MSC:
00-01
99-00

Keywords:
Social robots
Co-creativity
Child–robot interaction

ABSTRACT

Children's creativity is influenced by their social interactions with co-present peers during collaborative tasks, which digital pedagogical tools lack. Children are known to emulate social agents' behaviors. In this work, we explore how a social robot's *co-presence* and *creativity demonstration* influences children's creative expression during collaborative gameplay. Children played a digital drawing game that afforded *figural* creativity with a social robot (Jibo) as a peer-like playmate. To understand the effect of the robot's *co-presence* on children's creativity, we compared the creative expression of participants who played the game with Jibo, and a control group of participants who played in the absence of Jibo. No significant gains in creativity were observed. In a second study, we then enhanced the robot's behavior to express and model *figural* creativity during gameplay. To understand the effect of the robot's *creativity demonstration*, we compared the *figural* creativity of participants' drawings who played the game with Jibo while it modeled creative behaviors, with participants who played with a version of Jibo that did not. Children expressed significantly higher levels of *figural* creativity while interacting with the robot that modeled creative behavior through its drawings. We infer that while *embodied co-presence* was not a significant stimulant of creative expression, *creativity demonstration* led to heightened *figural* creativity. We suggest guidelines for designing robotic interactions to foster creative expression in children, highlighting the role of social emulation in influencing behavior.

1. Introduction

Creativity is defined by the ability to produce ideas with fluency, novelty, and value (Boden, 2004). Creativity in children has been shown to facilitate problem-solving, adaptability, self-expression, health, and increased learning benefits (Runco, 2014). Standardized creativity measures, such as the Torrance Test of Creative Thinking (TTCT) have previously demonstrated that as children progress from kindergarten to elementary school (ages 8–10), their creativity drops due to convergent and structured school curricula (Smith & Carlsson, 1985; Torrance, 1968; Williams, 1969). Fostering creative thinking for children is especially important in this era of Artificial Intelligence where repetitive and mechanistic jobs shift from humans to computers. Children learn creativity by emulating others in their environment, such as teachers and peers (Runco, 2014; Shaffer, 2006). Creative problem-solving in classrooms is often scaffolded by extrinsic factors, such as social interactions with co-located peers and teachers, collaboration with others, and the nature of the task itself (Dul & Ceylan, 2011; Kahn, Friedman, Severson, & Feldman, 2005; Runco, 2014). Creativity can also be fostered when activities are presented in "permissive and game-like fashion" (Williams, 1969).

Digital pedagogical tools, that are becoming increasingly common with the advent of remote learning and technologically assisted learning, often miss out on the creativity benefits of social interactions with co-located peers. Artificial embodied agents, such as social robots, have been used as effective pedagogical tools for young children, leading to both cognitive and affective gains (Belpaeme, Kennedy, Ramachandran, Scassellati, & Tanaka, 2018). Their ability to personalize learning, enhance engagement through expressive interactions and participate in collaborative learning activities as a peer situate social robots as an interesting learning intervention in classrooms. Among pedagogical tools, social robots have the unique ability to interact with children in a social-emotional way because of their ability to perceive and express emotion. Given how being co-located with peers and social interactions with others influence children's creativity, especially in collaborative tasks, in this work, we explore whether collaborative interactions with social robots can promote creative behavior in children. Specifically, we explore how the robots' *co-presence* and *creativity demonstration* during the task influence children's creativity. Creativity manifests in different forms

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https://doi.org/10.1016/j.ijcci.2020.100234
2212-8689/

namely, *verbal* and *figural*. We focus on children's *figural* creativity, which involves creative thinking in visual arts (e.g., drawing, painting, sculpting), since generating drawings is central to the interaction.

We designed a collaborative drawing interaction between a social robot and a child participant to understand the effects of human–robot interaction (HRI) on creativity outcomes. The game mechanics involved one player (child/robot) starting a drawing on a touch-screen tablet with one stroke and the other player completing it, before switching turns. We made use of the Sketch-RNN drawing model (Ha & Eck, 2017), trained on the QuickDraw dataset of drawings (Google, 0000), to generate drawing strokes that convert a starting stroke into a meaningful object. In order to understand the role of the robot player's co-presence on children's creativity, we ran a randomized controlled trial with 79 children in the 5–10 year-old age group. Participants were divided into two groups that were counterbalanced by their creativity scores (as measured by the TTCT test), age and gender. The control group of participants played the game in the absence of the robot (R-) and the second group of participants played with the robot (R+). We did not observe any significant differences in participants' *figural* creative expression (as measured by the Test for Creative Thinking–Drawing Production (TCT-DP)). We outline the study and results in Section 4.

Children are wired to learn with and from friendly others and socially emulate their behaviors and attitudes (Wood, Bornstein, & Bruner, 1989; Yando, Seitz, & Zigler, 1978). Research in Human–Robot Interaction (HRI) has previously shown how children socially emulate robots demonstrating different learning behaviors such as *verbal* creativity, curiosity, perseverance, growth-mindset and engagement (Ali, Moroso, & Breazeal, 2019; Gordon, Breazeal, & Engel, 2015; Park, Rosenberg-Kima, Rosenberg, Gordon, & Breazeal, 2017). They self identify more with these qualities after interacting with a robot peer that demonstrates them.

In order to understand the influence of *creativity demonstration* by the social robot on children's creativity, we carried out a second randomized controlled trial with 78 children in the 5–10 year-old age group. Participants were divided into two groups: The control group of participants interacted with the non-creative robot (C-) that did not exhibit creative behaviors and the second group of participants interacted with the creative robot (C+) exhibiting creative behaviors. Artificial creativity was enhanced by increasing the creativity of the drawings generated by improving the speed and accuracy of the drawing model. We observed that children who interacted with the creative robot scored significantly higher on *figural* creativity measures than children who interacted with the non-creative robot, and that the robot's creativity demonstration had a creativity contagion effect in participants. Hence, the robotic peer's behavioral design as a creative artist helped children be more creative. We outline the study and results in Section 5.

Finally, we suggest guidelines that inform the interaction pattern design of social embodied pedagogical tools to foster creativity in children.

2. Background

2.1. Creativity

Early researchers define creativity as the embodiment of thought in the form of external behavior, consisting of three characteristics: fluency, flexibility, and originality (Guilford, 1967). Modern researchers describe creativity as the process of generating artifacts or ideas, that are novel to the person, to the person's environment or to the world, that generate surprise, and that add value to the system (Boden, 2004; Kaufman & Sternberg, 2010). Creativity is often viewed in conjunction with divergent thinking or the potential for original thought (Runco, 2014). Sawyer understood creativity in the context of a group emergence where flow, collaboration, and improvisation processes take place (Sawyer et al., 2003). Creativity is expressed in different forms depending on the nature of the task and the medium of creative expression - *figural* creativity (e.g. drawing, painting, sketching) and *verbal* creativity (writing, storytelling, composition, discourse) (Guilford, 1957). In this work, we focus on *figural* creativity that is illustrated through a drawing task.

Even though creativity remains challenging to measure empirically, researchers have developed creativity assessment tasks that conceptualize and measure different forms of creativity (Cropley, 2001; Kahn et al., 2005; Khatena & Torrance, 1976; Meeker, Meeker, & Roid, 1985; Silvia, 2011; Torrance, 1972; Williams, 1993). We used the TTCT for assessing children's baseline creativity before all interactions and to divide them in balanced study groups. We used the TCT-DP to assess *figural* creativity from the drawings generated during the child–robot collaborative drawing task.

There are several external factors in classrooms, including their social interactions, that can contribute to fostering creativity in children. Researchers have identified the following "situational influences" of creativity: freedom, autonomy, good role models and resources, encouragement specifically for originality, freedom from criticism, and "norms in which innovation is prized and failure not fatal" (Amabile & Gryskiewicz, 1989; Witt & Beorkrem, 1989). Other creators in children's environment including teachers and peers also act as potential models for creativity (Runco, 2014). Creativity is also promoted when activities are presented in "permissive and game-like fashion" (Williams, 1969). Benefits of play-based learning approaches, and how it fosters child creativity have long been known (Papert, 1980). Isen, Daubman, and Nowicki (1987) demonstrate how positive affect facilitates creative problem solving, and games have been shown to induce positive affect in children. In our work, we utilize the knowledge of these determinants of creativity to design an effective creativity simulating interaction. The nature of the drawing game is collaborative, with the robot acting as a collaborative peer, as well as permissive and game-like. The robot exhibits creativity by generating novel artifacts (drawings).

Previous research has elaborated benefits of integrating creative skills into the educational institutions' curricula, and increased learning benefits (Runco, 2014). There have been several efforts towards creating Creativity Support Tools (CSTs), especially in the form of Graphical User Interfaces (Shneiderman et al., 2006). However, despite the social nature of creativity, not a lot of work has been done around utilizing artificial social agents as CSTs. Given that robots are increasingly being used in education, and have been shown to be effective learning tutors and companions (Belpaeme et al., 2018), this work aims to investigate the efficacy of peer-to-peer social interactions with robots as a Creativity Support Tool.

2.2. Emulation of social robots

Children learn from others through mechanisms of social emulation (Whiten, McGuigan, Marshall-Pescini, & Hopper, 2009), more specifically, end-state emulation (not to be confused with 'imitation'). Emulation is categorized as 'instances where children achieve common goals to those modeled, but do so by using idiosyncratic means that were never observed' and can be involuntary (Wood et al., 1989). Unlike imitation, it is only the end-state(s) of what the model has done that is copied. Copiers may reach this end in their own way, or might happen to choose the same method as the model. Social robots have previously been used as learning tools to foster positive learning behaviors, such as curiosity, growth-mindset, grit, persistence and attentiveness (Belpaeme et al., 2018; Gordon et al., 2015). Studies have looked at how children socially emulate a robot's learning behaviors, such as *verbal* creativity, empathy, curiosity or growth-mindset (Ali et al., 2019; Park et al., 2017; Zanatto, Patacchiola, Goslin, Thill, & Cangelosi, 2020). Gordon et al. demonstrated how children can socially emulate curiosity from a curious robot, and exhibit greater curiosity related behaviors such as free exploration, uncertainty seeking and question asking while interacting with a curious robot as compared to a non-curious robot (Ceha et al., 2019; Gordon et al., 2015). Park et al. demonstrated how a robot exhibiting growth-mindset can help foster a growth-mindset in young children (Park et al., 2017). Ali et al. demonstrated how children can emulate *verbal* creativity from a social robot (Ali et al., 2019). This establishes social emulation as an important mechanism of social learning that children seem to employ with peer-like social robots. In a similar theme, in this work, we aim to explore if a robot demonstrating *figural* creativity can help foster *figural* creativity as a learning behavior in young children.

2.3. Social robots and creativity

Robotic construction kits such as Lego Mindstorms, Bots Alive, Cozmo, Pop bots, Yolo (Alves-Oliveira, Arriaga, Paiva, & Hoffman, 2017; Anki, 1999; T. L. Group, 0000; Williams, 2018) have afforded creative expression in children. These works, however, use the robot as a toolkit that children use to create, rather than as an agent that children interact with. Fischer and colleagues (Fischer, 1993) embedded a software agent into an architectural design tool that offered critiquing statements to promote a reflective design practice. Jung et al. demonstrated how having a reflective conversation with an embodied artifact in the making positively influenced the learning process (Jung, Martelaro, Hoster, & Nass, 2014). Kahn et al. demonstrated that a robot's verbal and nonverbal behaviors such as question asking, positive reinforcement, challenging, and positive affect can help adults be engaged in a creative activity for longer times and come up with more creative ideas (Kahn et al., 2016).

In our previous work, we demonstrated how children model a social robot's *verbal* creativity during a verbal gameplay interaction (Ali et al., 2019). Participants played the Droodle creativity game with the social robot Jibo, which involved generating humorous creative titles for abstract figures. Participants that interacted with the creative robot exhibiting fluent ideas that are more novel and creative also generated significantly more fluent ideas and scored higher on novelty and creativity, as compared to participants that interacted with the non-creative robot. They demonstrated how children can emulate a social robot's artificial *verbal* creativity. Our work similarly explores whether children can emulate a social robot's artificial *figural* creativity.

In a previous study, Alves-Oliveira et al. (2019) studied the role of a robot, in comparison with a tablet, in stimulating creativity in humans in a collaborative drawing task. The social robot, controlled remotely in a Wizard of Oz manner, acted as a collaborative peer in a drawing activity, taking turns with the human, to complete their drawing. They found no significant effect of the robot's presence on children's creativity. Similarly, our work also utilizes a collaborative drawing interaction between children and a social robot, with the difference being that we made use of a fully autonomous child–robot interaction and drawing model. In our work, in addition to examining the role of embodied *co-presence* on children's *figural* creativity, we also look at *creativity demonstration* by the social robot as stimulant for children's *figural* creativity. Our goal was to encourage the social emulation of a peer-like robot's creative behaviors by young children.

3. System design

3.1. Robotic platform

For this study, we used Jibo, a socially expressive companion robot that has a three-axis body, a touch-screen based face, touch sensors on its head, two cameras to support face detection, text-to-text speech with speaker output, and automatic speech recognition (J. Inc., 0000). Jibo uses a web-socket to communicate with an Android tablet as a shared drawing surface. The game app's logic lives on an Android application, which communicates with Jibo over WiFi. We used Jibo since we wanted to build a completely autonomous collaborative robotic interaction, which was enabled by Jibo's ability to sense audio and receive commands from the tablet's interactions. Further, we wanted to make use of Jibo's bodily animations to express curiosity, joy and pride during the interactions. Jibo's child-friendly playful design made it ideal for the task. Even though Jibo does not have hands, the robot looked down at the screen, it's light lit up and moved its face along with the drawing, which when coupled with the robot's speech utterances, made it a believable experience. Participants believed this interaction since all participants answered the post-test questions about Jibo's ability to draw.

3.2. Collaborative drawing game

While iteratively playtesting with 19 children (age 6–10), we designed a collaborative game that aimed to foster *figural* creativity. In this game, Jibo takes the role of a fully autonomous playmate in a drawing game with a child using a touch screen tablet. The objective of the game is for the child and the robot to come up with a shared drawing. The tablet interface affords drawing using a finger, and makes it convenient to capture drawings' stroke data.

3.2.1. Game design

The gameplay is designed to be collaborative as collaboration is known to strongly influence and motivate creative thinking. The robot and the child take turns, building up on the other's drawing contributions, to complete the final figure. To begin, one player starts by drawing any initial doodle on the tablet and prompting the other player to convert the doodle into a meaningful goal object (such as a cat). See Fig. 1. The player can make multiple renditions of the drawing. The drawing goal is selected by the player from 55 preset model categories, that were chosen for age-appropriateness after consultation with a learning scientist. We initially designed the game to have five rounds of one minute each. However, in our playtests we observed that five rounds seemed too many for maintaining engagement, and one minute was too short to complete the drawing. We adapted the game to be played for three rounds of four minutes each (two minutes per player) before the game terminates (Fig. 2). This collaborative drawing task supports various aspects of creative thinking. First, players are encouraged to demonstrate divergent thinking, attempting to imagine how to build upon a starting doodle to complete a figure. It also supports fluency of thought, involving what Guilford terms as ideational fluency, i.e., the ability to rapidly produce ideas in succession. The game also supports associational fluency, i.e., the ability to generate artifacts to associate the starting prompts to the target object. The game mechanic also allows players to change the target object for the same prompt, or change the prompt for the same target object during the round. This requires the other player to demonstrate flexibility of thought, that is, come up with many different categories of ideas to solve the problem.

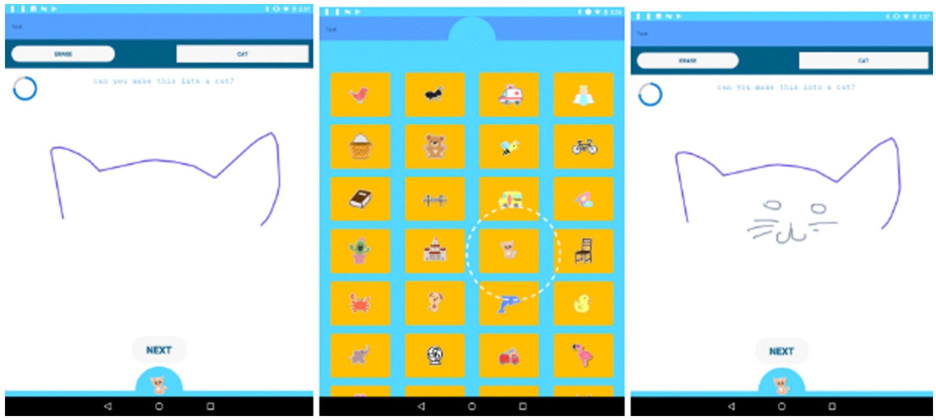

Fig. 1. Gameplay. Left to right: The child draws a starting prompt (cat ears). The child selects a target category *(cat)*. The robot converts the starting prompt into a *cat*.

Fig. 2. Turn-taking interaction of the collaborative drawing game. Left: The robot is completing the drawing prompt started by the child on the tablet interface as the child observes. Right: A child is completing the drawing prompt started by the robot.

We use the Test for Creative Thinking - Drawing Production (TCT-DP) as inspiration for designing the gameplay. The main difference is that the test starts from a fixed prompt and the subject converts the drawing into anything — whereas in our game, two players challenge each other to convert the initial stroke into an object that they chose for the other player. We chose this interaction to make it fun, collaborative and engaging for young children.

3.2.2. Generative drawing model

In order to design this collaborative drawing interaction, we needed an autonomous way to generate doodles that used the child's initial stroke as the starting point for the robot's drawing. We designed a fully autonomous interaction, where, during the robot's turn, the system generates drawings starting from the child's strokes to the final target doodle. We made use of Recurrent Neural Networks (RNNs) trained on human drawn images of 55 common everyday objects to generate these drawings. Taking inspiration from the Sketch-RNN model developed by Ha and Eck (2017), we trained a new model on crude human-drawn images representing 55 classes from the QuickDraw Dataset (Google, 0000), to generate unconditional images. Each class of QuickDraw is a dataset of 70K training samples, in addition to 2.5K validation samples, and 2.5K test samples. These 55 categories were chosen by a literacy expert based on what is age appropriate for children in the 5–10 year age group. The model is a Sequence-to-Sequence Variational Autoencoder (VAE), which takes a vector sketch as an input and outputs a latent vector. To predict endings of incomplete strokes, we used a method developed by Ha et al. where they use the decoder RNN as a standalone model to generate a sketch that is conditioned on previous points. The decoder RNN first decodes an incomplete sketch into a hidden state $\mathbf{h}$. Afterwards, they generate the remaining points of the sketch using $\mathbf{h}$ as the hidden state. These completed sketches can also be controlled in terms of their randomness by using the temperature variable τ.

The Sketch-RNN model runs on a local server on a computer in the vicinity of the tablet. The tablet sends and receives messages from the server which includes sending the starting prompt image and the category selected, and receiving the generated stroke image (Fig. 3). The RNN generates new drawings around the starting prompt and sends it back to the tablet. The tablet also sends the selected category to the robot, and the robot responds with the category name. The robot animations (looking down at the tablet and responding to the target category selected) provides the impression that the robot is driving the drawing process.

3.2.3. Robot behavior

The robot engages the child as a fully autonomous artistic peer that collaboratively makes drawings with the child. While the drawings are being created on the tablet screen, the robot logs information about which drawings are being drawn and when. The robot looks down upon the screen during its turn to draw and generates corresponding utterances. For instance, the robot says, *"Oh, an ant. I can make that drawing into an ant"*, *"Look at me go"* or *"Watch me convert your doodle into an ant"*. The robot also verbally engages the child by asking for feedback, *"What do you think about my drawings?"*, or *"Do you think that looks like an ant?"*. Additionally, the robot also displays the selected category icon on its face to demonstrate an association with the category chosen.

3.2.4. Interaction scenario

Introduction and tutorial. Children are first introduced to the robot by the experimenter as a peer that they would be playing the game with. The robot then engages in a short introduction to build rapport and introduce the game.

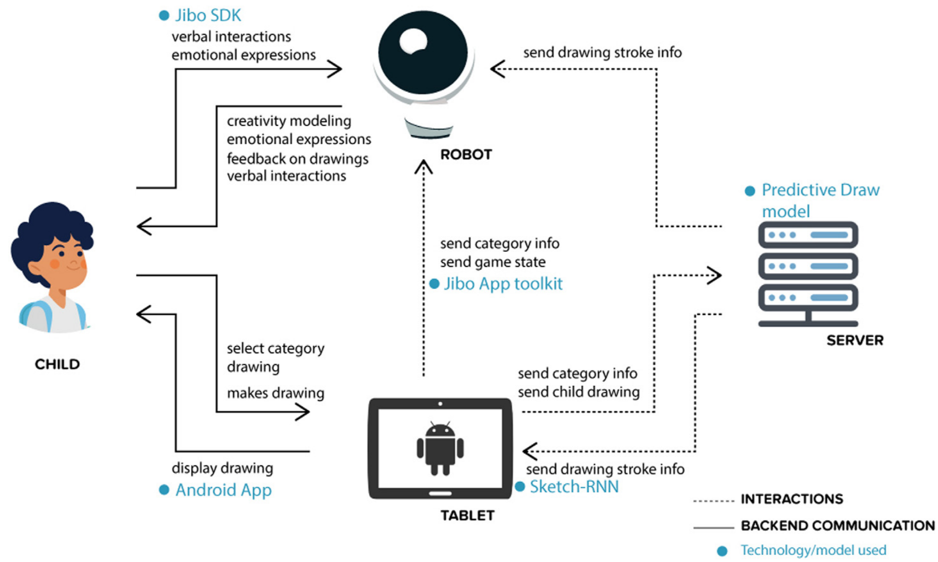

Fig. 3. System components. Children interact with the robot and the tablet verbally and for making drawings. The robot and the tablet communicate with the server to send category information and receive drawing stroke information.

ROBOT: Hi, my name is Jibo. I am a social robot. What is your name?
CHILD: Chasity
ROBOT: [*excitedly*] Hi Chasity. Welcome to *Drawing with Jibo*. We will be making some drawings together. Are you ready to begin?
CHILD: Yes
ROBOT: [*looking down at the tablet*] I will show you how to play the game. First, select a category from the menu below.

A menu drawer pops up. The child can then pull up the menu and select one of the categories (Fig. 1), for example, *ant*.

ROBOT: Oh, an ant. Now draw any shape on the screen, and watch me convert it to an ant.

The child draws a starting doodle, and the robot draws more strokes to convert it to an ant. The robot makes multiple ant drawings. Meanwhile the child can erase and draw a new starting prompt, or can select a separate category. The robot waits for two minutes to elapse before switching the turn.

ROBOT: Now, it's your turn. Can you convert this doodle into an elephant?

The child starts drawing. After initial playtesting, we included this tutorial in the first round to enable participants to understand the gameplay and require no experimenter intervention. Children interacted with the game smoothly barring the occasional crashing of the tablet due to connectivity problems. During initial playtesting, participants expressed confusion about why the robot was making multiple drawings. Adding speech utterances indicating intention such as, "Let me try another one", made that interaction smoother.

Gameplay. The game starts with the child drawing the prompt and selecting a category and the robot finishing the drawings. The robot draws multiple drawings for two minutes, before the turn switches.

ROBOT: Now, it's your turn to draw.

The robot draws a starting prompt and selects a category for the child to draw. Every time the child completes a drawing, the participant can press on the **Done** button. A screenshot of the drawing is taken with a sound and animation feedback. After the child completes a drawing, the robot provides feedback. The robot matches the drawing to the model category, and based on the confidence of image classification, it responds with a positive or neutral response. If the drawing has an 80% or higher match to the category, the robot responds with a positive feedback. Positive responses include, *"Good job"*, *"That was a good one"*, *"You are an artist"*, and *"That looks so much like an ant"*, and are accompanied by animations of *joy, excitement* or *happiness*. If the model has a match confidence of less than 80% with the category, the robot responds with a neutral feedback. Neutral responses include, *"Oh, was that an ant?"*, *"Let's try another one"*, or *"Do you want to try making another doodle?"*, and are accompanied by animations of *joy, confusion,* or *curiosity*. We deliberately chose not to include any negative feedback to avoid hindering creativity.

Both players play three rounds each, and then the robot terminates the game. In the initial playtests, some students were discontented with their drawings getting interrupted by turn changes, hence we introduced a timer displayed on the top left that indicated the amount of time left before the turn switches. The entire gameplay lasts about 15 to 20 min including the gap in turn switching and selecting categories.

ROBOT: Thank you for playing with me today. I had a lot of fun making these drawings. Did you have fun?
CHILD: Yes
ROBOT: I am going to go to sleep now, Chasity. See you next time. Bye!

4. Study design A: Effect of co-presence

To investigate whether a robot's embodied co-presence can positively influence children's creative expression, we designed a between-subjects randomized controlled trial.

4.1. Participants

79 participants of the 5–10 year-old age group were recruited for the study (40 female, 39 male). The average age of the participants was 7.35 (S.D. = 1.94). All participants and their guardians

Table 1
79 Participants participated in the study. Participants were divided into balanced groups based on their TTCT scores, gender and age.

Study groups	n	TTCT scores	Gender	Age
R-	38	41.66 ± 6.01	F = 20, M = 18	7.61 ± 1.92
R+	41	42.91 ± 5.16	F = 20, M = 21	7.09 ± 1.96

signed a consent to participate and for audio and video data collection. The recruitment materials, study protocol and data collection protocol were reviewed and approved by the Institute Research Board at MIT.

4.2. Pretest

All 79 participants were administered the TTCT as a part of the pretest activity at least one week before the study. The TTCT is a paper based evaluation that consists of 2 sets of assessment activities: *verbal* creativity test and *figural* creativity test. The purpose of conducting the TTCT was to drive a quasi-random assignment into groups such that their creativity scores are balanced across two conditions.

4.3. Study conditions

Participants were divided into two study condition groups, as shown in Table 1. We performed repeated random clustering until the differences in the participants' pretest creativity scores, age and gender were minimal across the two groups:

Tablet-only condition (R-). In the control group, participants played the game on a tablet screen with no co-present peer. The tablet would act as the other player, taking turns to complete the participants' drawings. In order to control for the time and nature of interaction, the dialogs designed for the robot during gameplay, turn-shifting, and drawing, were expressed using audio prompts voiced from the tablet's speaker.

Robot condition (R+). In the robot condition, participants played the game with Jibo, taking turns to complete each other's drawings.

The gameplay in the two conditions was exactly identical, and the parameters of the drawing model used were identical to the tablet-only condition. The temperature parameter τ of the generative model to be the 0.8, the tablet screen displayed drawings at the frame-rate of 30 fps.

4.4. Hypothesis

Children who played the collaborative drawing game with the robot (R) will exhibit higher levels of creativity in their own drawings than children that played in the absence of the robot (R-).

4.5. Post-test

In order to understand children's perceptions of the robot's creativity and peer-like influence during the interactions, we administered a post-test questionnaire consisting of a likert scale and yes/no questions:

- Q1 : Do you think Jibo (or the tablet) helped you finish the drawings? (yes/no)
- Q2 : Do you think Jibo (or the tablet) is a better artist than you? (yes/no)
- Q3 : On a scale of 1 to 5, how good an artist do you think Jibo is? (5 being highest, 1 being lowest)
- Q4 : On a scale of 1 to 5, how much fun did you have playing the game today? (5 being highest, 1 being lowest)

The word *artist* was used in the post-test questionnaire instead of *creative*, since *creative* was a difficult term for young children to understand.

4.6. Data collection and measures

4.6.1. Data collection

We used the following sensor setup or logging method to collect gameplay data:

- Tablet action logs:
  - All drawings drawn by the child in three rounds.
  - All drawings drawn by the robot in three rounds.
  - Time taken for each drawing.
- Overhead GoPro camera:
  - Video of the interaction (birds-eye view).
  - Audio of the interaction.
  - Researchers conducting the study recorded the post-test interview responses of the participants.

4.6.2. Creativity measures

We used the TCT-DP test to analyze the creativity of the drawings. Our collaborative drawing game is different from the TCT-DP since it is not open-ended like the test, in order to keep it collaborative and game-like and it is conducted on a tablet instead of paper. Since there are no available tests for less open-ended drawing tasks, we make use of a modified TCT-DP to measure the drawings' creativity, where we exclude some metrics of TCT-DP, that are not relevant to this task, as outlined below:

- Continuations (Cn): Any use, continuation or extension of the given figural fragments.
- Completion (Cm): Any additions, completions, complements, supplements made to the used, continued or extended figural fragments.
- New elements (Ne): Any new figure, symbol or element.
- Connections made with a line (Cl) between one figural fragment or figure or another.
- Connections made to produce a theme (Cth): Any figure contributing to a compositional theme or "Gestalt". *Since the theme was predefined in our task, we excluded this parameter.*
- Boundary breaking that is fragment dependent (Bfd): Any use, continuation or extension of the "small open square" located outside the square frame.
- Boundary breaking that is fragment independent (Bfi).
- Perspective (Pe): Any breaking away from two-dimensionality. *We exclude this parameter since we used a tablet for the drawing task.*
- Humor and affectivity (Hu): Any drawing which elicits a humorous response, shows affection, emotion, or strong expressive power.
- Unconventionality, a (Uc, a): Any manipulation of the material such as paper. *We exclude this parameter since we used a tablet.*
- Unconventionality, b (Uc, b): Any surrealistic, fictional and/or abstract elements or drawings.
- Unconventionality, c (Uc, c): Any usage of symbols or signs.
- Unconventionality, d (Uc, d): Unconventional use of given fragments.
- Speed (Sp): A breakdown of points, beyond a certain score-limit, according to the time spent on the drawing production.

Table 2
TCT-DP scores of participants. Unpaired t-test revealed that the study condition did not have a significant effect on participants' *figural* creativity.

Group classifications	Percentile	TCT-DP	Modified TCT-DP
Far below average	0–10	0–17	0–13
Below average	11–25	18–29	14–23
Average	26–75	30–52	24–40
Above average	76–90	53–57	41–45
Far above average	91–97.5	58–62	46–48
Extremely high above average	≥2.5%	63–64	49–50

Table 3
TCT-DP scores of participants. Unpaired t-test revealed that the study condition did not have a significant effect on participants' *figural* creativity.

Study groups	n	TCT-DP scores
Tablet-only (R-)	33	27.75 ± 11.11
Robot (R+)	34	32.88 ± 9.64
p		$p = 0.06$

Group classifications of the regular TCT-DP (14 parameters) and our modified TCT-DP (11 parameters) are shown in Table 2

We recruited three coders, who were blind to the study's hypothesis and the participants' study condition, to review the drawings generated in the child's turn and rate them. We conducted an inter-coder reliability test to validate the reliability of coding and found a high correlation of 0.82. We used these scores for calculating the TCT-DP measures of the drawings. After excluding the data points of participants that did not complete the experiment and participant whose data points were not recorded, we had a total of 67 drawings (R- = 33, R+ = 34) that we ran analysis on.

4.6.3. Condition analysis

Children's creativity was calculated in terms of the TCT-DP measures. We conducted an independent-samples unpaired t-test between the two conditions to compare the creativity scores.

Post-test analysis. We compared the post-test self report measures using unpaired t-tests to reveal differences in different study conditions.

4.7. Results

We ran Shapiro–Wilk (S–W) test to check for normality and Levene's test to check for equal variance for the dependent variables. We failed to reject Shapiro–Wilk's null hypothesis ($p>0.05$) that the data is normally distributed, and Levene's null hypothesis ($p>.05$), hence we conclude that there is insufficient evidence to claim that the variances are not equal. Hence, we conducted unpaired t-test comparing creativity scores of drawings generated by participants in the R- control condition and R+ condition-test, as measured by the TCT-DP test. The unpaired t-test revealed that participants that interacted with the robot (R+) generated drawings that scored higher on creativity (M = 32.88, S.D = 9.64), than the control group (R-), however this difference was not significant (M = 27.75, S.D = 11.11), t(63) = 1.99, $p = 0.06$ (Fig. 4). The results are shows in Table 3.

Hence, the data did not provide enough evidence to support our hypothesis that the robot's embodied co-presence had an effect on children's *figural* creativity during a collaborative drawing game.

In the post-test questionnaire, we learned children's insights about the game interaction, and how they perceive the robot's ability to create and help them create.

Q1. 29.04% participants in the R+ condition reported that Jibo was helpful in finishing their drawings, and 27.03% participants in the R- condition reported that the tablet was helpful.

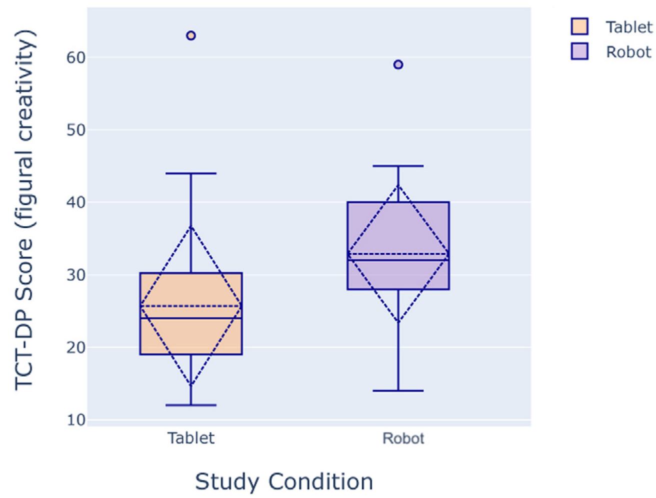

Fig. 4. Participants that interacted with the robot scored higher on the TCT-DP test compared to the control group, however, this difference was not significant. $p = 0.06$.

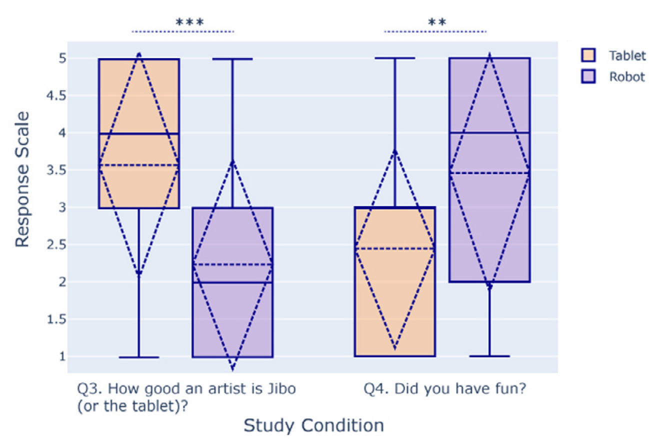

Fig. 5. Q3 & Q4 result. Participants interacting with the tablet perceived the tablet to be more creative and the interaction to be less fun than participants interacting with the robot.

Q2. Participants in the R- condition were more likely to confirm that the tablet was a better artist than them (57.89%) than in the R+ condition (27.03%), even though they use the same drawing model.

Q3. Using an unpaired t-test, we found that participants in the R- condition rated the tablet a significantly better artist than participants in the R+ condition rated the robot: R+ (M = 2.24, S.D. = 1.42) and R- (M = 3.58, S.D = 1.53); t(65) = 1.99, $p =$ .0002. (Fig. 5). Participants perceived the robot to be less creative than the tablet, even though they were using the same drawing model.

Q4. Participants in the R- reported to find the interaction significantly less fun that participants in the R+ condition: R+ (M = 2.44, S.D. = 1.34) and R- (M = 3.46, S.D = 1.61); t(70) = 1.99, $p = .004$. (Fig. 5). Even though the presence of the robot had no influence on children's creativity, it made the interaction more fun.

5. Study design B: Effect of creativity demonstration

To investigate whether a robot's own creative behavior can positively influence children's creative expression, we designed a

Table 4
78 participants participated in the study. Participants were divided into balanced groups based on their TTCT scores, gender and age.

Study groups	n	TTCT scores	Gender	Age
C-	41	42.91 ± 5.16	F = 20, M = 21	7.09 ± 1.96
C+	37	43.33 ± 6.30	F = 14 M = 23	7.89 ± 1.91

between-subjects randomized controlled trial. We also report the results of this Study Design B: Effects of Creativity Demonstration in our previous work (Ali, Park, & Breazeal, 2020).

5.1. Participants

78 participants of the 5–10 year-old age group were recruited for the study (34 female, 44 male). The average age of the participants was 7.53 (S.D. = 1.93). 3 participants did not complete the interaction, and the results for 8 participants was not recorded accurately. Hence, only data from 67 participants was used for the analysis.

5.2. Pretest

78 participants were recruited for the study (using the same mechanisms as Study A) and were administered the TTCT. We conducted quasi-random assignment into groups such that their creativity scores are balanced across two conditions.

5.3. Study conditions

Participants were divided into two study condition groups, as shown in Table 4. We performed repeated random clustering until the differences in the participants' pretest creativity scores, age and gender were minimal across the two groups:

Non-creative robot condition (C-). In the non-creative robot condition, the drawing model was configured at its default settings, where it did not produce highly creative doodles. The temperature parameter τ of the generative model was 0.8, which led to some randomness in the drawings with a lower model match to the category that the child selects. The frame-rate of rendering in the default model was set to 30 fps. The model also periodically selected an incorrect category to make the model seem less of an expert artist.

Creative robot condition (C+). We adapted the system in this condition to produce more creative drawings as defined by the TCT-DP figural creativity test. Creativity was exhibited through the quality of drawings, in terms of drawing creativity metrics described in the TCT-DP test: Continuations (Cn), Completion (Cm), New elements (Ne), Connections made with a line (Cl) between one figural fragment or figure or another, Connections made to produce a theme (Cth) and speed of drawing (Sp) (Urban, 2005). During the robot's turn, we adjusted the temperature variable of the drawing model τ (Cm, Cl, Cth) to be 0.2 to reduce the randomness in drawing. Further, we increased the speed of drawing to 60 fps. The robot always drew true to the selected category. This led to higher quality drawings with a better model match to the category that the child selects.

The length and number of interactions were controlled for across the two conditions.

Table 5
Drawings generated by the C+ model were significantly more creative than the C- as measured by TCT-DP.

Model condition	n	TCT-DP score
Non-creative model (C-)	15	20.33 ± 7.86
Creative model (C+)	12	28.91 ± 6.69
p		$p = 0.0064^{**}$

Validation of study conditions. While we adjusted model variables to create more creative drawings in the C+ condition as compared to the C- condition, we analyzed the actual drawings produced during the robot's turn to verify if that was indeed the case. Specifically, we compared the TCT-DP scores of 12 robot drawings generated by the creative model and 15 robot drawings by the non-creative model using the same starting prompts. An unpaired t-test revealed that the model type had a significant effect ($p < 0.01^{**}$) on the generated drawing's creativity scores (Table 5). In terms of absolute TCT-DP scores, the C- drawings lied in the far below average to below average range, with the mean in the below average range, and the C+ drawings lied in the average to above average range, with the mean in the average range. Hence, we validated that the manipulation of the model to adjust temperature, pick categories and adjust drawing speed led to a change in the drawing's creativity as intended per conditions.

5.4. Hypothesis

Children who played the collaborative drawing game with the creative robot (C+) will exhibit higher levels of creativity in their own drawings than children that played with the non-creative robot (C-).

5.5. Post-test

In order to understand children's perceptions of the robot's creativity and peer-like influence during the interactions, we administered a post-test questionnaires:

- Q1 : Do you think that Jibo helped you finish the drawings? (yes/no)
- Q2 : Do you think that Jibo's talking was helpful for your drawings? (yes/no)
- Q3 : Do you think that Jibo is a better artist than you? (yes/no)
- Q4 : On a scale of 1 to 5, how good an artist do you think Jibo is? (5 being highest, 1 being lowest)
- Q5 : On a scale of 1 to 5, how much fun did you have playing the game today? (5 being highest, 1 being lowest)

The data collection, creativity measures and analysis were identical with Study A (Section 4.6).

5.6. Results

We ran Shapiro–Wilk (S–W) test to check for normality and Levene's test to check for equal variance for the dependent variables. We failed to reject Shapiro–Wilk's null hypothesis (p>0.05) that the data is normally distributed, and Levene's null hypothesis (p>.05), hence we conclude that there is insufficient evidence to claim that the variances are not equal. Hence, we conducted unpaired t-test comparing creativity scores of drawings generated by participants in the C- control condition and C+ condition-test, as measured by the TCT-DP test. The unpaired t-test revealed that participants that interacted with the creative robot (C+) generated drawings that scored significantly higher on creativity (M = 42.27, S.D = 14.63), than participants that interacted with

Table 6
TCT-DP scores of participants. Unpaired t-test revealed that the study condition had a significant effect on participants' *figural* creativity.

Study groups	n	TCT-DP scores
Non-Creative robot (C-)	34	32.88 ± 9.64
Creative robot (C+)	33	42.27 ± 14.3
p		$p = .0023^{**}$

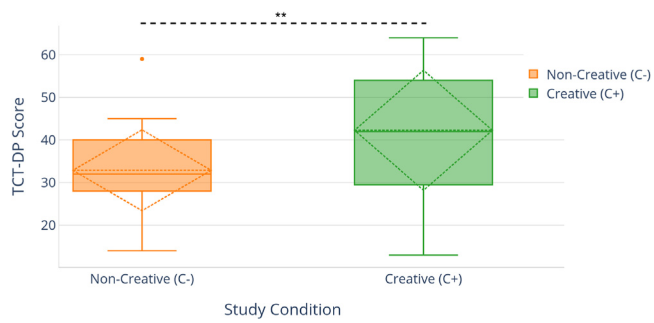

Fig. 6. Participants that interacted with the creative robot scored significantly higher on the TCT-DP test compared to the non-creative robot.

the non-creative robot (C-) (M = 32.88, S.D = 9.64), t(65) = 1.99, p = .0023 (Fig. 6). The results are shows in Table 6.

Hence, we verified our hypothesis that children emulate the robot's expressed *figural* creativity during a collaborative drawing game.

In the post-test questionnaire, we learned children's insights about the game interaction, and how they perceive the robot's ability to create and help them create.

Q1. More participants in the C+ condition reported that Jibo was helpful in finishing their drawings (65.85%) than participants in the non-creative robot condition (27.03%). Participants viewed the more creative peer as the more helpful one.

Q2. More participants in the C+ condition thought of the robot's talking to be helpful (C+ = 60.98%, C- = 24.32%). One participant in the C+ condition said "Yes, it was helpful to me when I was asking him what to do and he did it right away, he didn't need other reminders to do it, it is really easy to give him what to do or say". Another participant explained "Yes, he said good to my drawings".

Q3. Participants in the creative robot condition were more likely to confirm that the robot was better than them (C+ = 39.02%, C- = 27.02%). However, a majority of participants in both conditions did not think that the robot was a better artist than they were.

Q4. With unpaired t-test, we found significant difference in how the participants perceived the robot to be an artist, children in C+ condition rating the robot higher than children in the C- condition: C+ (M = 2.97, S.D. = 1.51) and C- (M = 2.24, S.D = 1.12); t(65) = 1.99, p = .03. (Fig. 7). We can conclude that on average, participants were able to successfully perceive the higher quality of drawings of the C+ robot.

Q5. With unpaired t-test, we found no significant difference in whether participants in one condition found the game more or less fun in the C+ condition: C+ (M = 3.66, S.D. = 1.42) and C- (M = 3.46, S.D = 1.61); t(65) = 1.99, p = .56. (Fig. 7). The robot's creative expression led to no change in children's perception of the interaction as more fun.

6. Discussion

In this work, we designed a novel fully autonomous child–robot drawing interaction aimed at fostering *figural* creativity in young children. Through the first study, we demonstrated how a social robot's co-presence has no influence on children's *figural* creativity. The robot's co-presence did make the interaction significantly more fun for children. However, it was unexpected that children perceived the tablet to be more creative than the robot, even though both the study conditions used the same drawing model. One possible reason for this difference could be that children have a higher expectation from the robot's creativity skill due to its embodiment, and hence rate it lower. Further studies are required to rationalize this finding.

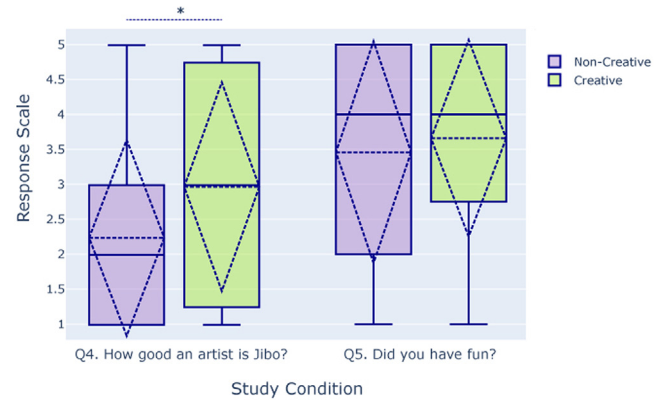

Fig. 7. Q4 & Q5 result. Participants in the C+ condition perceived the robot to be more creative than participants in the C- condition (p = .0023**).

Through the second study, we demonstrate how artificial *figural* creativity exhibited by a social robot in a collaborative drawing task is emulated by children, and in turn led to their drawings being more creative as measured by the TCT-DP test of measuring creativity in drawings. A game-based interaction where the robot is positioned as a collaborative creative peer is a contagion in children's creativity. Participants that interacted with the creative robot drew significantly more creative drawings than participants who interacted with the non-creative robot. We infer that a social robot's expression of high creativity is socially emulated by young children and facilitates creative gains in children. Participants not only exhibited higher creativity, but also perceived the robot as a better artist.

Similar to previous work by Alves-Oliveira et al. (2019), the present work did not find evidence that the embodied co-presence of the robot influenced children's creativity. We deduce that it is not just the embodiment of the collaborative peer, but also the artificial creativity exhibited by it, that can foster creativity in children. Children's creative learning is influenced by their social interactions with others and peers often act as models for creative behavior. From prior work in HRI, we learn that children socially emulate robotic peers' learning behaviors such as *verbal* creativity, curiosity and growth-mindset. Through a randomized controlled trial, we demonstrated that when the robot exhibited high *figural* creativity in a collaborative task, it in turn stimulated creativity in children.

Digital pedagogical tools are becoming commonplace, especially with the emergence of both remote learning and technologically equipped classrooms. While many of these tools effectively translate the cognitive learning benefits of classroom learning to digital media, they lose out on the behavioral learning benefits of classrooms, such as curiosity, empathy, resilience, persistence and creativity, which they often learn from emulating their peers. Research in HRI has demonstrated how children are likely to socially emulate agents that they see as more peer-like and more social. Social robots have already found a place in education as learning companions, and have been shown to have cognitive and affective learning benefits for young children due to their ability to facilitate personalized learning (a common struggle in large classroom sizes), and higher engagement (than non-embodied interfaces) (Belpaeme et al., 2018). Through this work, we demonstrate the efficacy of social robotic agents to promote learning behaviors such as creativity through themselves demonstrating creative behaviors in collaborative tasks. Designers of pedagogical tool must consider integrating social robots as models of target behaviors (figural creativity in this case), that can be emulated by young children.

6.1. Design guidelines

Through our experience of iterative design and evaluation of a child–robot interaction that successfully promoted creative expression in children, we formulated guidelines to design creativity stimulating interactions, with a special focus on the design of game-based interactions and social robots as creativity support tools. These guidelines can benefit researchers, game designers and developers aiming to foster creative expression in children.

6.1.1. Nature of the task

Creativity expressed during a task is influenced by several factors, one of which is the nature of the task itself. We adopted a game-based task, since playful interactions provide space for creative thinking. The task revolved around the creation of new artifacts (drawings). Open-endedness, such as not having 'one correct drawing' helps promote divergent thinking. Generative machine learning techniques enable the creation of a unique artifact in each attempt. It is essential to determine the optimum difficulty and length of the games. We consulted learning scientists to determine object word categories that students are familiar with, and adapted the robot speech to use child-friendly vocabulary. We increased the length of the drawing rounds since participants in the playtests found it difficult to draw quickly.

6.1.2. Collaboration with social agents

Among external factors that influence creativity, collaboration is one of the most prominent ones. While designing child–robot interactions, we must ensure that the interactions are collaborative in nature and the robot acts as a collaborative peer instead of a competitive one. This is especially imperative for creativity, since competition and evaluation hinder creativity in children.

6.1.3. Social emulation to foster creativity

We learned that children can learn creativity while interacting with a social robot expressing creative behaviors. While designing social robots as pedagogical tools, we must ensure that we program their behaviors to model artificial creativity.

6.1.4. Voicing intent

In order to make the connect between the robot and the artifact it creates, as well as to make the interaction seem natural, we found voicing the robot's intention to be a beneficial technique. For instance, when the robot starts drawing, it looks down and makes utterances such as *"Look at me go."*.

6.1.5. Rapport building

Through iterative prototyping, we observed that children find the interaction fun and interesting if we begin by building a peer-like rapport and establishing common ground. This involved personalized and encouraging dialog such as, 'what is your name?' or 'I like dancing, what do you like to do?' or 'Are you excited to play this game with me today?'

7. Conclusion and future work

In this work, we have designed a fully autonomous child–robot interaction to stimulate children's creativity. We have studied the effects of a social robot's *co-presence* and *artificial creativity demonstration* of a social robot on children's creativity. We found that when the robot demonstrated *figural* creativity, it in turn stimulated creativity in children, where children who interacted with the robot generating more creative drawings also generated more creative drawings. Since social robots have known to be effective pedagogical and are already being used in classroom settings as learning peers and personalized tutors, it is imperative to critically think about how these robots' behaviors can influence children's learning behaviors, such as creative thinking. We demonstrate how a robot's expressed creativity can be emulated by young children during gameplay.

Advances in generative Machine Learning have opened up the avenues of various forms of artificial artistic expressions. Embodied AI agents have the potential to use generative networks to express forms of creativity through generating media such as drawing, poetry, art styles, patterns, physical body movements, etc. This work opens up the opportunity to explore how these different forms of artificial creativity can be embedded in agents that children interact with, and help them be more creative.

It is important to note that while social robots are not the only way to provide this creativity support through behavioral modeling, they are a compelling way. This technology is an affordable way to scale the support, and amplify and augment personalized social interaction. In this work, we study how artificial *creativity demonstration* by the social robot can simulate creativity in children by comparing the interaction with a non-creative robot. In future work, we would explore how the robot's social interactions such as reflective speech, positive reinforcement, and expressed affect influences children's creativity and affect.

One limitation of the work is that in the collaborative drawing game, we designed a game-like digital implementation of the TCT-DP, which is a paper-based test. In using a digital UI, there are some parameters of creativity defined by the TCT-DP that the game does not allow, such as *boundary breaking*, that are more affordable on paper. In order to mitigate this limitation, in future work, we can use physical co-creative drawing interactions, such as, in previous work by Lin, Guo, Chen, Yao, and Ying (2020). There also exists an opportunity to develop better generative drawing models to draw more realistic and complex drawings. It is also important to note that in this study, we recruited students from 5–10 year old age group, which includes a large developmental gap. While we observed no significant correlations of *figural* creativity scores with students' age, it would be beneficial to run the study with narrower age ranges to derive more age-specific insights.

In this study, we compared the creative expression of two groups, however, in order to understand how their baseline creativity was affected, we must conduct a pre- and post-test independent of the robot to determine changes in absolute creativity. Further, long-term effects on children's creativity by using such an intervention are not known. Lastly, this work is also limited to narrow constructs of *figural* creativity, and only consisted 55 categories of drawings of simple objects. Creativity encompasses a much wider array of behaviors that can be explored using other interactions and mediums such as music generation or construction games. As a part of our future work, we aim to expand this study to other models for creativity and evaluate whether children emulate other forms of the social robot's creativity, such as *constructional* creativity. We also aim to design more open ended co-creation tasks that allow unhindered creative expression. It is also interesting to investigate how the effect of a social robot's creative scaffolding in one task transfers to another task without the robot being present, and hence evaluate the long-term effect on children's creative thinking.

8. Selection and participation

78 participants of the 5–10 year-old age group were recruited for the study (34 female, 44 male). The average age of the participants was 7.53 (S.D. = 1.93). The subjects were recruited as a part of the East Somerville Secondary School after-school activities program at the public schools in Somerville, Massachusetts, USA. All participants and their guardians signed a consent to participate and for audio and video data collection. After participants signed the assent form and before beginning the study the participants were again reminded in easy terms how the data is being collected. All participants were briefed about the robot and the task that they will be performing by the researcher. They were also informed that they can ask the researcher any questions they have about the game or the robot and may choose to stop participating in the study at any time. The recruitment materials, study protocol and data collection protocol were reviewed and approved by the Institute Research Board at the Massachusetts Institute of Technology (MIT). 3 participants did not complete the interaction, and the results for 8 participants was not recorded accurately. Hence, only data from 67 participants was used for the analysis.

Declaration of competing interest

The authors declare the following financial interests/personal relationships which may be considered as potential competing interests: Massachusetts Institute of Technology Jibo Inc.

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Update

International Journal of Child-Computer Interaction

Volume 41, Issue , September 2024, Page

DOI: https://doi.org/10.1016/j.ijcci.2024.100675

International Journal of Child-Computer Interaction 41 (2024) 100675

Contents lists available at ScienceDirect

International Journal of Child-Computer Interaction

journal homepage: www.elsevier.com/locate/ijcci

Erratum regarding missing declaration of competing interest statements in previously published articles

Declaration of Competing Interest statements were not included in the published version of the following articles that appeared in previous issues of International Journal of Child-Computer Interaction.

The appropriate Declaration/Competing Interest statements, provided by the Authors, are included below.

"On power and participation: Reflections from design with developmentally diverse children" (International Journal of Child-Computer Interaction 27 (2020) 100241) https://doi.org/10.1016/j.ijcci.2020.100241. Declaration of competing interest: The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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DOIs of original article: https://doi.org/10.1016/j.ijcci.2020.100234, https://doi.org/10.1016/j.ijcci.2020.100220, https://doi.org/10.1016/j.ijcci.2020.100166, https://doi.org/10.1016/j.ijcci.2012.08.001, https://doi.org/10.1016/j.ijcci.2020.100240, https://doi.org/10.1016/j.ijcci.2020.100210, https://doi.org/10.1016/j.ijcci.2020.100207, https://doi.org/10.1016/j.ijcci.2019.100144, https://doi.org/10.1016/j.ijcci.2019.100142, https://doi.org/10.1016/j.ijcci.2020.100214, https://doi.org/10.1016/j.ijcci.2020.100169, https://doi.org/10.1016/j.ijcci.2018.04.002, https://doi.org/10.1016/j.ijcci.2020.100232, https://doi.org/10.1016/j.ijcci.2020.100218, https://doi.org/10.1016/j.ijcci.2020.100182, https://doi.org/10.1016/j.ijcci.2020.100172, https://doi.org/10.1016/j.ijcci.2020.100167, https://doi.org/10.1016/j.ijcci.2019.100152, https://doi.org/10.1016/j.ijcci.2020.100241, https://doi.org/10.1016/j.ijcci.2020.100186, https://doi.org/10.1016/j.ijcci.2019.100151, https://doi.org/10.1016/j.ijcci.2020.100216, https://doi.org/10.1016/j.ijcci.2020.100188, https://doi.org/10.1016/j.ijcci.2019.100146, https://doi.org/10.1016/j.ijcci.2020.100239, https://doi.org/10.1016/j.ijcci.2020.100212, https://doi.org/10.1016/j.ijcci.2020.100201, https://doi.org/10.1016/j.ijcci.2020.100199, https://doi.org/10.1016/j.ijcci.2019.100164, https://doi.org/10.1016/j.ijcci.2019.100148.

https://doi.org/10.1016/j.ijcci.2024.100675

Available online 18 July 2024
2212-8689/

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